Sahaj Bhadja

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Summary

New-grad test engineer with 2+ years building **Python automation, simulation, and validation systems**; experienced translating requirements into test cases, configuring test environments, executing integration tests, diagnosing failures, documenting results, and improving coverage through CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Developed **Python validation and reconciliation scripts** for 5,000+ monthly workflows, checking state transitions, malformed data, and integration failures while achieving 99%+ production reliability through automated CI/CD releases.
- Diagnosed performance and data defects using structured logs, error states, and targeted test scenarios, reducing dashboard latency from 1.2 seconds to 350 milliseconds and improving concurrent processing throughput 3x.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Derived and executed **integration test cases** for PDF, DOCX, image, and ZIP requirements, reproducing failures across multiple AI and OCR components and implementing validation, fallback handling, and persistent results.
- Wrote Python automation for multithreaded batch testing, processing 1,000+ files per run and 10,000+ daily requests while maintaining stable execution under peak load and reducing turnaround time 60%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **closed-loop simulation and evaluation pipelines** across 10,000 autonomous-driving episodes, integrating map geometry, temporal trajectories, and system constraints to improve long-horizon planning 18% and behavioral stability 15%.
- Designed repeatable validation scenarios, measured system performance, and investigated off-road and lane-following failures, documenting technical findings and collaborating with researchers to refine configurations and iteratively improve model behavior.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Configured **VM-based test environments** across Windows, Linux, and macOS, automating nightly builds, installation, integration testing, and reporting while validating 20+ tests and improving overall cross-platform release coverage.
- Supported root-cause analysis using failure logs, coverage reports, and isolated component tests, collaborating with Microsoft and GitHub engineers to troubleshoot integration defects and deliver controlled production fixes.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Executed **system integration and end-to-end tests** across eight components and 2M+ synchronized records, validating authentication, resource allocation, notifications, WebSocket updates, and Azure SQL behavior before staged deployment.
- Used Sentry, Azure Monitor, and Kibana to reproduce and isolate application, interface, and data failures, documenting results for cross-functional teams and shipping verified Docker releases through Jenkins 50% faster.

Skills

Languages: Python, C++, C, Bash, Java, JavaScript, SQL
Validation: Test cases, test automation, integration testing, functional testing, regression testing, fault isolation
Systems: Systems engineering, simulation, test environments, Linux, Windows, TCP/IP, REST APIs, WebSockets
Tools: PyTest, Mockito, Selenium, Git, Jenkins, GitHub Actions, Docker, Sentry, Azure Monitor, Kibana
Engineering: Requirements analysis, root-cause analysis, metrics, coverage, technical reporting, AI-assisted workflows

Projects

CPU Scheduling Simulator Live | *C++, Algorithms, Functional Testing*

- Implemented and tested **six C++ scheduling algorithms** with interactive controls, Gantt visualization, edge-case handling, and dynamic metrics across 1,000+ process combinations, reducing average turnaround time 25% and debugging time 20%.

Linguistix | *Python, AWS, Docker, CI/CD*

- Developed and deployed a **containerized Python validation service** on AWS ECS/Fargate with CodePipeline, automated quality checks, and repeatable CI/CD releases, achieving 87% accuracy and production-grade operational reliability at scale.

CaptionCraft Live | *Python, PyTorch, Computer Vision*

- Designed a **repeatable model-validation benchmark** across four vision architectures, 10,000 images, and 100,000 captions, analyzing failures and test results to improve accuracy from 81% to 96% while building inference APIs.